NGUYEN QUANG DUY

AI ENGINEER | LLM APPLICATIONS, MCP TOOLING & ML PLATFORMS

Ho Chi Minh City, Vietnam | 0902577827 | nquangd1510@gmail.com

Portfolio: duytruyenthong.github.io/Alinguyen-portfolio/

TECHNICAL SKILLS

AI applications and retrieval: Ollama; AI-assisted personal projects using Groq's OpenAI-compatible API, prompt/context design, JSON-schema outputs, citation validation, LangChain Core/LCEL, MCP and SQLite FTS5/BM25 lexical retrieval.

ML platforms and evaluation: ML training/serving pipelines, on-prem GPU infrastructure, reproducible experiments, model packaging/versioning and automated deployment; defined LLM grounding and prompt-injection evaluation fixtures with AI assistance for a personal project.

Data and programming: Python, SQL, JavaScript, Node.js and React.js; Databricks reporting pipelines, DuckDB, PostgreSQL, Pandas and scikit-learn. SQL joins/CTEs/window functions and dashboard building with AI assistance.

Cloud and operations: AWS EC2/VPC/S3/Lambda/CloudFront; GitHub Actions, Bash, Ansible, Nginx, IaC and security scanning; BullMQ, RabbitMQ, Elasticsearch and Redis; Linux, Prometheus, Grafana and CloudWatch Agent.

MarTech and automation: Meta, Google, TikTok, Shopee, AppLovin and Pinterest ads; GA4, Google Tag Manager and AppsFlyer; n8n and Power Automate. Advertising budgets: USD 2,000-10,000/month per product.

Working approach: Systems thinking and problem framing. Claude and Codex: 10,000 hours of combined use.

PROFILE

AI engineer combining LLM background processing, ML platform tooling and cloud delivery with data-pipeline validation and AI-assisted application projects.

PROFESSIONAL EXPERIENCE

KiteMetric | AI Engineer

Jan 2025 - Sep 2026 | Hybrid

- Built BullMQ background workers for Ollama text generation and ffmpeg video processing, coordinating these workloads through queued jobs.
- Used deterministic job IDs to limit duplicate processing and implemented startup reconciliation for background jobs.

ML platform and lifecycle responsibilities

- Built ML pipelines and supporting platform tooling for model training, experimentation and production serving.
- Designed and operated on-prem GPU infrastructure and a self-service platform for reproducible experiments.
- Developed a model lifecycle framework for packaging, versioning and automated deployment.
- Implemented infrastructure as code, CI/CD security scanning and end-to-end observability across ML pipelines and serving systems.

Application delivery and cloud operations

- Built asset-management CRUD and recursive folder traversal; used S3 pre-signed URLs for direct file transfers to object storage.
- Integrated AWS Lambda/Sharp processing, Redis URL caching and CloudFront asset delivery.
- Implemented Elasticsearch search over post archives with typo-tolerant fuzzy matching and keyword highlighting.

PROFESSIONAL EXPERIENCE - CONTINUED

KiteMetric | AI Engineer - continued

Jan 2025 - Sep 2026 | Hybrid

- Built GitHub Actions/Bash CI/CD for rolling EC2 deployments, versioned S3 artifacts and rollback across staging and production.
- Provisioned Ubuntu/Debian servers and AWS VPC, EC2, load balancers and Auto Scaling Groups; used Ansible for repeatable configuration and CloudWatch Agent for monitoring.
- Developed RabbitMQ ingestion for enterprise Windows log events, separating webhook handling from database work.
- Configured Prometheus/Grafana infrastructure and application dashboards with email alerts.
- Configured Nginx reverse proxying and traffic balancing across backend services.

iTechWX | Microsoft Technical Support Specialist

Apr 2025 - Feb 2026 | Night shift

- Provided enterprise Microsoft 365 support across OneDrive, SharePoint, Exchange Online and Teams, investigating incidents affecting collaboration services and daily business operations.
- Troubleshot technical issues for enterprise partners involving Microsoft Intune, intranet and server environments.
- Reproduced reported behavior, checked relevant service configurations and tested diagnostic hypotheses to narrow incident causes and document technical evidence. Investigated authentication, licensing, synchronization and permissions incidents, linking reported symptoms to the affected Microsoft 365 service and its configuration.
- Analyzed email-delivery incidents with Message Trace and MX/DNS verification; used Microsoft Admin Center and PowerShell to test service-configuration hypotheses and document findings.
- Managed enterprise incidents from intake through customer communication, technical documentation and resolution tracking; coordinated product-level escalations with Microsoft engineering teams. Verified service behavior following resolution and reviewed recurring issue patterns; contributed reusable troubleshooting knowledge for consistent handling of future cases.

NINA Web Service | Solution Consultant

Dec 2023 - Dec 2024

- Conducted client discovery to capture business objectives, operational constraints and user needs, using these requirements to define the problems the proposed website should address.
- Translated business requirements into website architecture, user-experience needs and expected functionality, making the intended solution behavior explicit for implementation.

NINA Web Service | Data-platform contribution

During NINA employment; project dates not specified

- Implemented a Databricks transformation pipeline to clean raw data-lake inputs and produce structured tables for reporting. Validated source-to-report transformations by reconciling input/output row counts and checking transformed records for duplicates and missing values.
- Cross-checked sample output records against source data to verify reporting consistency. Reviewed pipeline job logs and reran the Databricks pipeline, comparing repeated outputs to confirm consistent reporting results.

SELECTED PYTHON / SQL DATA PROJECT

Paid professor database project | Python / SQL

Project-based work began in March 2022

- Paid professor database project: Python and SQL database-building work processing approximately one billion records in total. Project-based work began in March 2022.

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EARLIER DEVELOPMENT EXPERIENCE

NINA Web Service | Web Developer

Jun 2023 - Dec 2023

- Developed responsive React.js/JavaScript interfaces and maintained existing web applications, with attention to user-facing functionality, usability and maintainable implementation. Built front-end components and supporting application functionality; investigated defects, tested changes and integrated updates into existing applications.
- Made iterative improvements to application functionality and interfaces as part of the development process. Prototyped a knowledge-based support chatbot using retrieval-augmented generation and AI-agent concepts for user assistance.

Amazing Tech | Junior Front-End Developer

Sep 2022 - Jan 2023

- Contributed front-end refactoring and responsive UI implementation alongside senior developers.
- Supported debugging and code-maintainability improvements under senior-developer guidance.

SELECTED DATA AND AI PROJECTS

STOS | Sales Traffic Operating System

AI-assisted personal product project

- Built a sales-page product with templates, a page studio and inquiry capture.
- Built an Insights workflow for page views, actions and leads; AI suggestions remain planned.

KKBox | Customer Churn Prediction Baseline

AI-assisted personal project | Local, within-cohort holdout

- Developed a Python/DuckDB feature pipeline aggregating transactions and listening events through 28 February 2017, with source-coverage audits and snapshot-date filtering.
- Built a logistic-regression baseline with median imputation, standard scaling and train-only preprocessing on a fixed stratified 75%/25% train/holdout split.
- Added synthetic future-event exclusion tests and recorded input hashes and aggregate audit counts for reproducibility.

Career Agent Kit | AI-assisted retrieval and orchestration

Personal engineering project

- Build a temporary SQLite FTS5 index in Python over confirmed source quotes, rank matches with BM25 and retain source IDs and hashes in bounded context packets. Queries use parameterized SQL; retrieval remains lexical, without embeddings or a vector database.
- Expose local MCP tools for career context and observed measurements, keeping retrieval separate from account access and application-execution authority.
- Composed optional LCEL RunnableLambda stages for retrieval, a host drafting callback and revalidation; copied context before the callback and limited concurrency to one.

EDUCATION AND SCHOLARSHIP

Languages: Vietnamese; English IELTS 6.5; German A2; Chinese HSK 2.

Bachelor of Computer Science

Vietnamese-German University | 2019 - 2024

- Selected coursework: Linear Algebra, Statistics and Discrete Mathematics.
- DAAD Study-Stay Scholarship, 2021-2022 - not taken due to COVID-19 circumstances.

Bachelor of Science | Computer Science and Engineering

Frankfurt University of Applied Sciences | Dual-degree qualification